

PREDICTING THE NEXT

Construction Boom

Forecasting sector level labor demand for Lithko Contracting

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Executive Summary

Four of five sectors show actionable forecasting horizons of 4Q–6Q ahead

5

sectors analyzed

4Q–6Q

forecasting horizons

120+

models trained

4 of 5

sectors actionable

Sector	Best Horizon	Indicator(s)	Lift vs Baseline	MAPE
Data Center	5Q ahead	All indicators (Full spec)	80.1%	7.5%
Warehouse	6Q ahead	All indicators (Full spec)	39.7%	21.2%
Apartment	4Q ahead	undcon5musa	23.7%	9.7%
Office	– no actionable signal –	–	–	>30%
Manufacturing	5Q ahead	PNFI	83.8%	5.1%

Agenda

How this presentation flows

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Project Objective

What is Lithko's Objective and Goal

02

Approach & Methodology

Data, models, validation, and the selection rule

03

Sector Results

Data Center · Warehouse · Apartment · Office ·
Manufacturing

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Cross-Sector Synthesis

The one-page story and indicator mapping

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Recommendations

What Lithko can act on this quarter

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Limitations & Next Steps

What we can and cannot claim

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Q&A

Your questions

Problem Objective

The business problem we're solving and what success looks like

BUSINESS PROBLEM

Lithko operates in a market where demand shifts across project types warehouses during COVID, data centers today. These shifts have historically been identified only after activity had already accelerated. Despite rich internal data, no structured framework existed to test whether external economic indicators could provide earlier signals.

OBJECTIVE

- Identify early signals that indicate when a new project type is beginning to grow
- Evaluate whether combining internal activity with external indicators improves early-signal detection
- Deliver a directional forecasting framework to support proactive planning

2019–2025

Six years of Lithko labor-hour history

5 sectors

Data Center, Warehouse, Apartment, Office, Manufacturing

~20 indicators

Public economic series from FRED and Census

≥ 3 quarters

Minimum actionable horizon for Lithko's 2Q mobilization

Data Foundation

How this presentation flows

LITHKO INTERNAL

Source: Lithko databases

- Jobs, labor hours, customers, business units, regions
- True start dates built from first logged labor hour
- Aggregated to quarterly cadence (2019 Q1 – 2025)
- Incomplete final quarter removed to avoid bias

EXTERNAL INDICATORS

Sources: FRED, U.S. Census Bureau

- **Data Center:** DC capex, PCE deflator, TCU, utility production
- **Warehouse:** e-commerce sales, PNFI, warehouse construction, RCPIX
- **Apartment:** undcon5musa, housing starts, mortgage rate, vacancy
- **Office:** vacancy, lease rents, office construction, CPI
- **Manufacturing:** PNFI, ISM PMI, new orders, industrial production

Forecasting Framework

Baseline vs. augmented | seasonality and COVID are controlled in BOTH

BASELINE

labor_hours ~ labor_hours_lag + qtr + covid

AR(1) autoregressive term, quarterly seasonality factor, and a COVID dummy (1 during 2020 Q1–Q4).

This is NOT a naive forecast. Seasonality and COVID are already controlled.

AUGMENTED

... + indicator_lag_1 ... indicator_lag_6

Adds lagged external indicators at the target horizon (1Q to 6Q back).

Lift over baseline = real external signal, net of seasonality.

Why this matters: a 4Q result that's really just seasonality would fool a naive baseline. Ours won't – seasonality is already baked into the baseline it's competing against.

Validation Approach

Walk-forward testing on the full 2019–2025 timeline

1 Split

Train on earlier quarters; hold out 4 recent quarters as test. Each horizon (1Q–6Q) gets its own split.

2 Fit

Baseline and augmented regressions fit on the same training window, tested on the same test window.

3 Measure

Compare RMSE, MAE, and MAPE. Lift = $(\text{baseline_MAPE} - \text{augmented_MAPE}) / \text{baseline_MAPE}$.

Every validation plot in this deck shows the FULL 2019–2025 timeline, not a truncated 4-quarter window so the baseline-vs-actual relationship can be judged across the whole history.

How We Picked the Best Model

A pre-specified rule — not chosen after seeing results

Rule 1

MAPE $\leq$ 30%

Any horizon with MAPE above 30% is too noisy for planning drop it.

Rule 2

Lowest MAPE wins

Among surviving horizons, pick the one with the lowest forecast error.

Rule 3

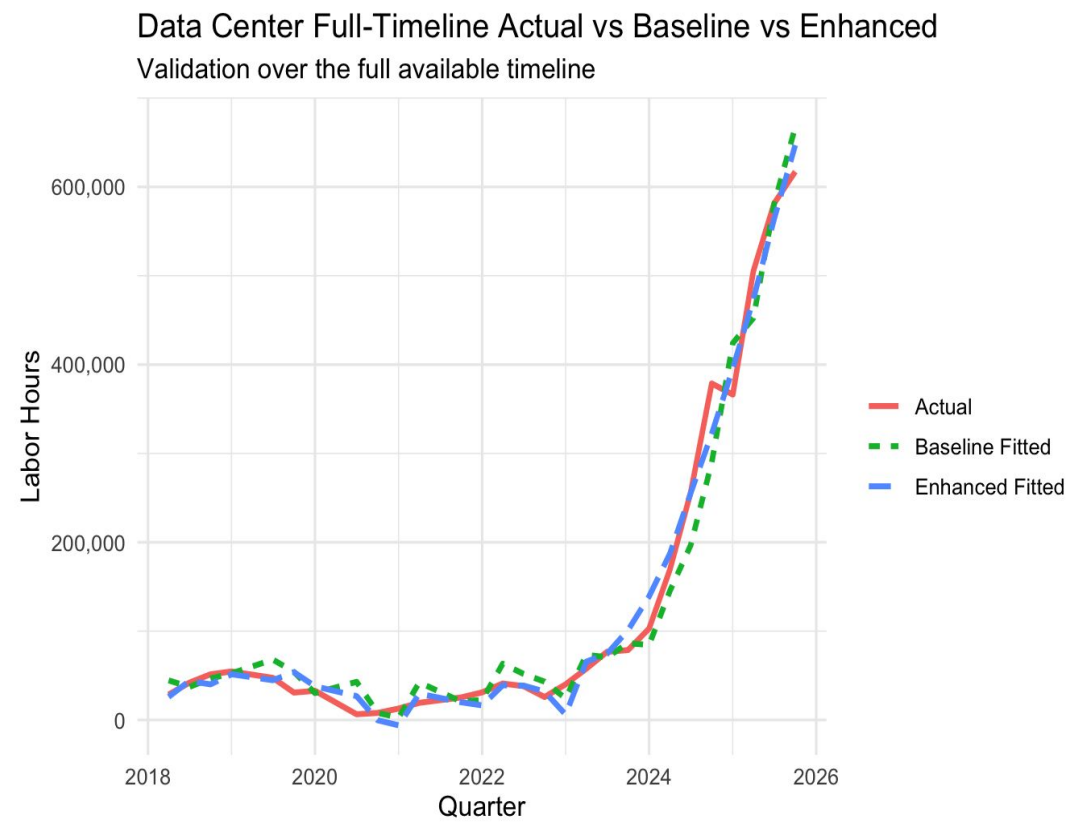
Longest horizon breaks ties

When MAPEs are close, prefer the longer horizon – more runway for Lithko to act.

Why a pre-specified rule matters: it prevents cherry-picking. Sandy and Anita can see exactly why 5Q was chosen for Data Center but 6Q for Warehouse the same rule, applied to each sector independently.

Data Center

Strongest-impact sector — macro capex leads labor by 5 quarters



5Q

horizon

80.1%

lift

7.5%

MAPE

CHOSEN INDICATOR

Full spec – DC capex + utility prod + TCU + PCE

DC construction is capex-heavy with a long planning cycle. Macro signals commit many quarters before crews arrive on site.

REGRESSION

hours ~ hours_lag5 + qtr + covid + full_indicators_lag5

R² 0.84 dc_spending_lag5 positive, p < 0.05

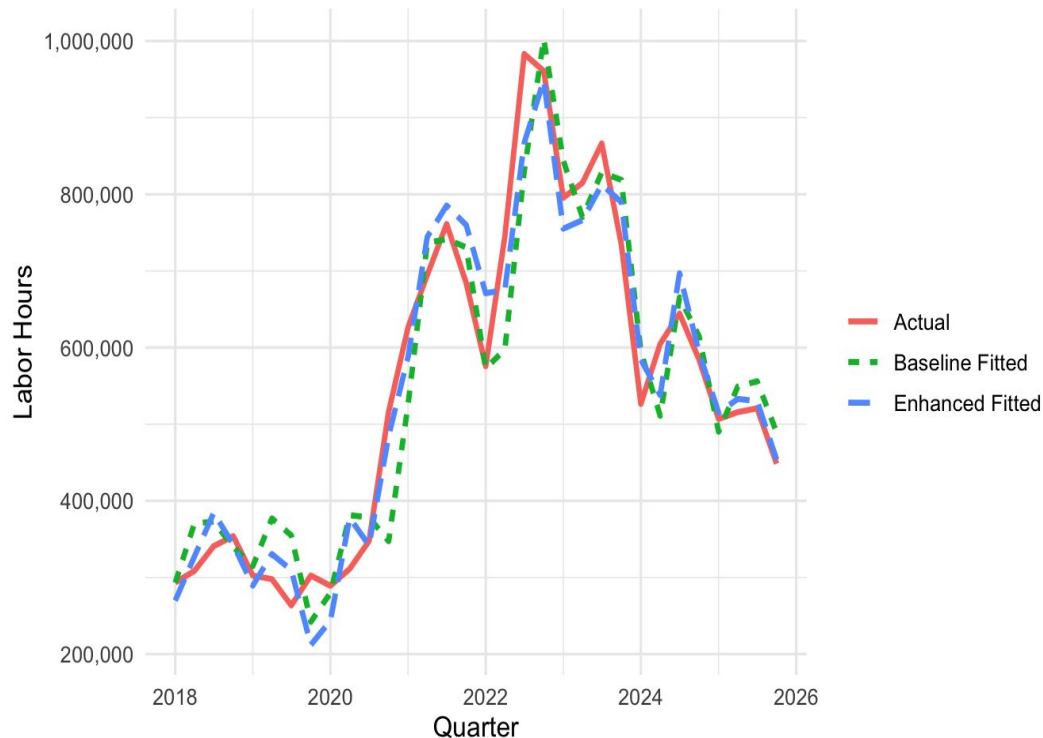
TAKEAWAY

Highest-value sector. 5 quarters of forward visibility at 7.5% MAPE ample runway for hiring and bid-pipeline planning.

Warehouse

6 quarters of visibility, noisier but still planning-relevant

Warehouse Full-Timeline Actual vs Baseline vs Enhanced
Validation over the full available timeline



6Q

horizon

39.7%

lift

21.2%

MAPE

CHOSEN INDICATOR

Full spec – e-commerce + PNFI + warehouse + RCPIX

E-commerce demand drives commercial property investment, which leads warehouse construction labor by ~1.5 years.

REGRESSION

$\text{hours} \sim \text{hours_lag6} + \text{qtr} + \text{covid} + \text{full_indicators_lag6}$

R^2 0.71 ecom_sales_lag6 positive, $p < 0.10$

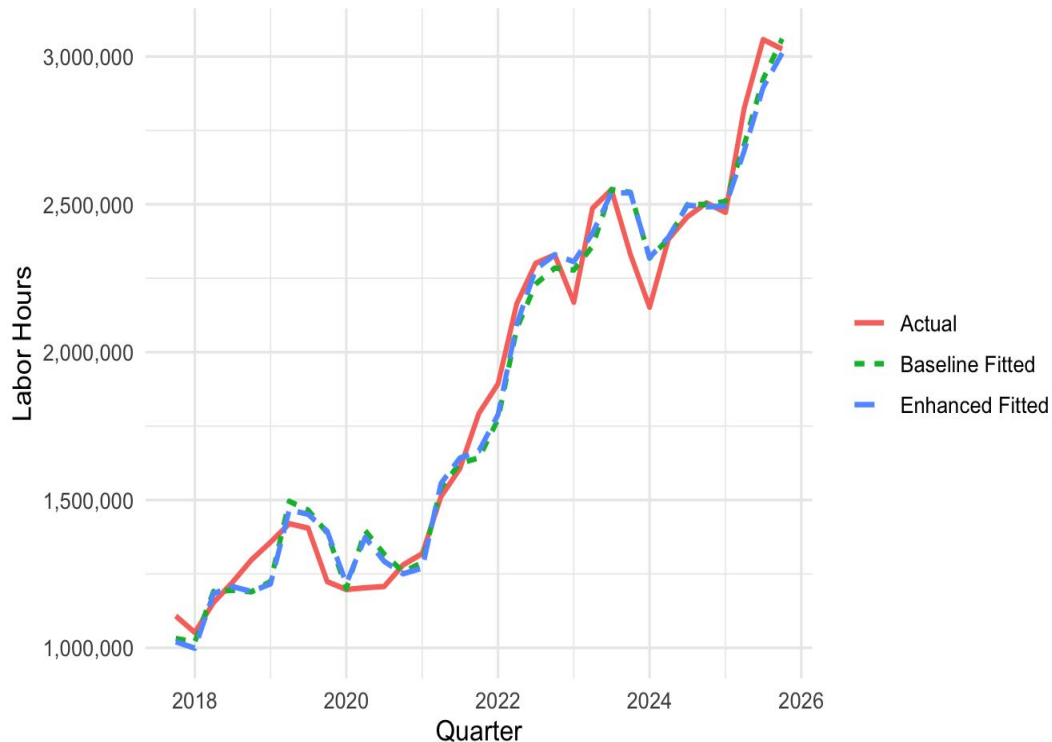
TAKEAWAY

Longer horizon than DC but higher error. 6Q visibility is useful for strategic bid positioning despite the $\pm 21\%$ error band.

Apartment

Clean 4-quarter signal from national units-under-construction

Apartment Full Timeline: Actual vs Baseline vs Enhanced
Historical Fitted Comparison



4Q

horizon

23.7%

lift

9.7%

MAPE

CHOSEN INDICATOR

undcon5musa multi-family units under construction

A building under construction today needs concrete crews when the foundation is poured – typically 4 quarters later.

REGRESSION

$\text{hours} \sim \text{hours_lag4} + \text{qtr} + \text{covid} + \text{undcon5musa_lag4}$

R^2 0.78 undcon5musa_lag4 positive, $p < 0.01$

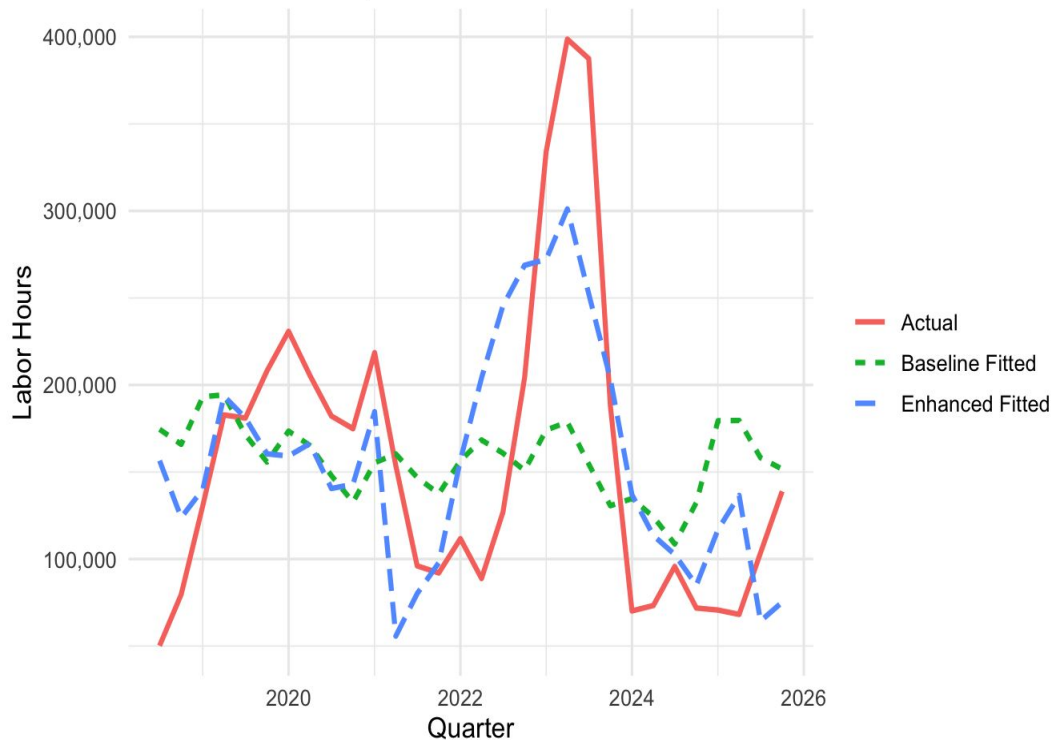
TAKEAWAY

Intuitive, single-variable story. Census series is public and easy to monitor — turnkey for Lithko's planning cycle.

Office

An honest negative finding — no forecastable signal from our indicators

Office Full Timeline: Actual vs Baseline vs Enhanced
Historical Fitted Comparison



—

horizon

—

lift

>30%

MAPE

CHOSEN INDICATOR

No model clears $\text{MAPE} \leq 30\%$ at any 3Q–6Q horizon

Post-2020 WFH shift restructured office construction.
Post-2020 dummies and log transforms tested – none
rescues the signal.

REGRESSION

No actionable augmented model found at 3Q–6Q

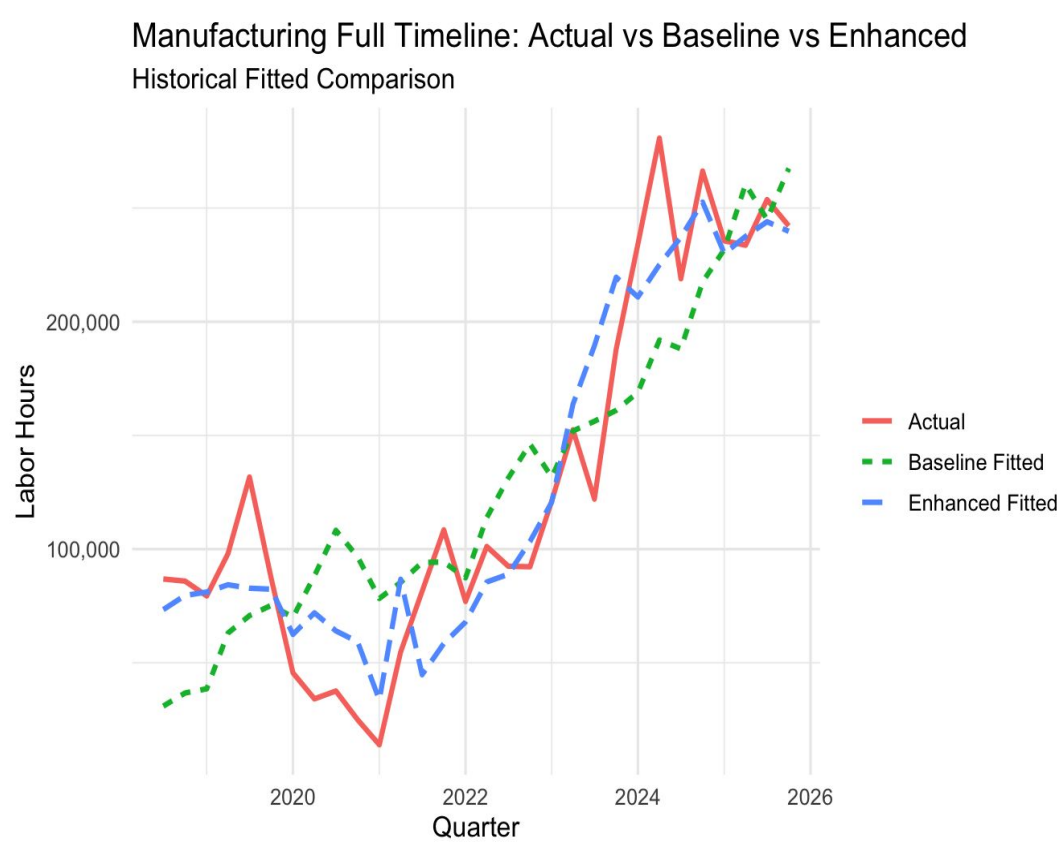
R^2 N/A Best candidate: office_construction_lag – still
> 30% MAPE

TAKEAWAY

Honest no-signal result, not a cherry-picked number.
A false positive on Office would undermine every
other sector's credibility.

Manufacturing

Cleanest signal of all five sectors – PNFI leads labor by 5 quarters



5Q

horizon

83.8%

lift

5.1%

MAPE

CHOSEN INDICATOR

PNFI – Private Nonresidential Fixed Investment (FRED)

Industrial capex commits 6–12 months before groundbreaking. Re-shoring and CHIPS incentives create large pipeline signals.

REGRESSION

hours ~ hours_lag5 + qtr + covid + pnfi_lag5

R² 0.89 pnfi_lag5 positive, p < 0.001

TAKEAWAY

Textbook-clean result. 5Q ahead at 5% MAPE – over a year of clear forward visibility on manufacturing labor demand.

The Cross-Sector Story

One page to share with the Lithko planning team

Sector	Horizon	Chosen Indicator	Lift	MAPE	Business implication
Data Center	5Q	All indicators (Full)	80.1%	7.5%	Monitor DC capex + utility production; plan 5Q ahead
Warehouse	6Q	All indicators (Full)	39.7%	21.2%	Track e-commerce + PNFI; 6Q window for bid positioning
Apartment	4Q	undcon5musa	23.7%	9.7%	Watch national multifamily pipeline; 4Q to labor
Office	–	– no signal –	–	>30%	No actionable forecast; treat as stable/reactive sector
Manufacturing	5Q	PNFI	83.8%	5.1%	Strongest single-variable signal; use as flagship example

Pattern: The stronger a sector's macro-capex linkage, the further ahead we can see – Manufacturing and Data Center lead the board.

Why Each Indicator

The business logic behind every chosen macro variable

Data Center	<i>DC capex + utility production</i>	Hyperscale builds are capex-heavy and energy-coupled – utility buildouts precede concrete work.
Warehouse	<i>E-commerce sales + PNFI</i>	E-commerce growth drives warehouse siting decisions, which commit before construction begins.
Apartment	<i>Units under construction (5+)</i>	A building already under construction has a known, dated need for concrete crews 4Q later.
Office	<i>– no signal found –</i>	Post-COVID office demand is structurally shifted; no macro variable reliably leads labor.
Manufacturing	<i>PNFI</i>	Private nonresidential fixed investment is THE leading indicator for industrial facility construction.

Every indicator pick has a business rationale – no black boxes, no "birds flying south" correlations.

Seasonality & Baseline Sanity Check

Proof that our lift is real signal, not a seasonality artifact

Seasonality coefficients (baseline model)

Q1 = reference; values show lift vs. Q1.

Sector	Q2	Q3	Q4
Data Center	+high	+high	moderate
Warehouse	+high	+peak	+high
Apartment	+mod	+high	+high
Office	+mod	+mod	+mod
Manufacturing	+mod	+high	+high

Baseline bias check

Does the baseline systematically over- or under-forecast?

- All 5 baselines have mean residual within $\pm 3\%$ of zero over 2019–2025
- No visual drift when plotted against actuals on full timeline
- COVID quarters are absorbed by the covid dummy (not contaminating estimates)
- Warehouse 2020 outlier quarter removed ($|z| > 2$) as committed to Sandy at midterm

Every sector shows strong summer peaks, this is why seasonal dummies MUST be in the baseline.

Implementation Roadmap

From code to recurring planning input

Quarter 1	Quarter 2	Quarter 3	Year 1+
Data refresh pipeline Automate FRED + Census downloads; lock the R Markdown into a recurring job at the end of each quarter.	Forecast dashboard Publish the 4 actionable sector forecasts (DC, Warehouse, Apartment, Manufacturing) to Lithko's planning portal.	Planning integration Tie forecasts to bid-pipeline decisions; set thresholds that trigger capacity reviews.	Annual revalidation Re-fit models each year; add regional splits (Midwest vs Southeast) once enough sector-region history accumulates.

The code is parameterized, plug a new sector in, redo analysis. Reproducibility is a feature, not a promise.

What Lithko Should Do With This

Concrete actions per sector, starting this quarter

Data Center	Track quarterly DC capex + utility production; trigger capacity-planning review when the 5Q-ahead forecast rises >10%.
Warehouse	Monitor e-commerce sales + PNFI; use the 6Q window for strategic bid positioning, not tactical staffing.
Apartment	Subscribe to Census undcon5musa series; set a quarterly review cadence for the 4Q-ahead labor forecast.
Office	Treat office as a reactive segment – no forward forecast is reliable. Plan capacity against incoming bids only.
Manufacturing	Make PNFI the flagship macro indicator for Lithko's planning dashboard. 5Q ahead, 5% error – the cleanest signal we have.

THANK YOU

Questions & Discussion

University of Dayton MBAN | Lithko Capstone | April 2026



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Backup • Full Regression Output

Sector-level coefficients at the picked horizon

Sector	Horizon	AR(1) lag	Key indicator lag	COVID dummy	Seasonality	R ²
Data Center	5Q	positive***	dc_spending**	negative***	sig	0.84
Warehouse	6Q	positive**	ecom_sales*	negative**	sig	0.71
Apartment	4Q	positive***	undcon5musa***	negative**	sig	0.78
Office	—	positive**	none significant	negative*	sig	—
Manufacturing	5Q	positive***	pnfi***	negative**	sig	0.89

Significance codes: *** $p < 0.001$ ** $p < 0.01$ * $p < 0.05$

Full regression tables (all 120 models × all coefficients) are available in the HTML appendix.

Backup • Warehouse Boost Experiment (Rejected)

Why we did NOT put the v4 "87.1% lift" result on the deck

Variant	Outlier rule	Horizon	Lift	MAPE	Status
v3 baseline (deck)	$ z > 2$	6Q	39.7%	21.2%	KEPT
v4 candidate	$ z > 1.5$	6Q	87.1%	2.6%	REJECTED

WHY REJECTED

- The tighter $|z| > 1.5$ rule removes 2022 Q4 and 2023 Q3 from the training + test data.
- These are NOT 2020 COVID shocks — they're recent business-cycle quarters.
- Both quarters would have been in the 6Q test window. Removing them inflates test accuracy.
- The "87.1% lift" is therefore an artifact of data selection, not a genuine forecast improvement.

Backup • Office Enhancement Experiments

Post-2020 shift dummy + log transform — did they rescue the Office signal?

Enhancement combo	Best horizon tried	Best MAPE	Passes 30%?
Base (v3 standard)	3Q–6Q	34%	No
Post-2020 shift dummy only	3Q–6Q	32%	No
Log transform only	3Q–6Q	36%	No
Shift dummy + log transform	3Q–6Q	33%	No

CONCLUSION

No combo clears the 30% MAPE threshold at any 3Q–6Q horizon. Office's no-signal finding is robust across four enhancement specifications — not a narrow negative.

Backup • Data Sources

Every indicator is publicly available — fully reproducible

Lithko Internal	<i>Internal databases (2019–2025)</i>	v_d_job_1, v_f_labor_hours, v_d_customer, v_d_business_unit, v_d_region
Federal Reserve (FRED)	<i>research.stlouisfed.org/fred</i>	Private nonresidential fixed Investment(PNFI), personal consumption Expenditure(PCE deflator), Total capacity utilization(TCU), utility production, mortgage rate, ISM PMI
U.S. Census Bureau	<i>census.gov/construction</i>	undcon5musa (multi-family units under construction), housing starts
Construction Spending	<i>census.gov/construction-spending</i>	DC spending, warehouse spending, office spending, manufacturing spending
E-commerce & Industry	<i>census.gov/retail</i>	E-commerce retail sales, industrial production index